

Conditional Probability

A, B are two events (outcomes) of a random experiment

Prob of A given B

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



$P(A|B) = 1/2$
 $P(A) = 1/4$
 $P(B) = 1/2$
 $P(A \cap B) = 1/4$

Ex. A = first die = 3
 B = second die = 5
 C = sum dice = 6

$P(A|B) = P(A) = 1/6$
 $P(A|C) = ??$
 A = spades
 B = black
 C = red

Test 1000 people for cancer

500 cancer → 450 +
 50 -
 500 don't have cancer → 100 +
 400 -

$P(A|B) = .9$
 $?? P(B|A) ??$
 Bayes' Thm Inference

Think about what the probability of A given B means: the probability of finding **A after B has happened**.

Here, don't get confused that we are finding the probability of A alone. We are finding the probability of A given that B has already happened.

So, if we have two dependent events, A and B, which have a common area representing $A \cap B$, then $P(A \cap B)$ is the probability that both A and B happen.

The probability of A after B has happened is the probability of $A \cap B$, divided by $P(B)$, which represents the effect of B having already happened.

So, this is how we get the formula:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Problem: cancer patients – 500 have cancer and other 500 do not have cancer,

but the machine says 450 +ve, 50 -ve instead of 500+. So, 50 of the results are wrong and the machine says 100+ even they don't have cancer,

so, let says $p(a)$ is having cancer, which is $500/1000 = 50\%$

$p(b)$ is device saying +ve = $550/1000 = 55\%$

given that if b already happened which says 500 have cancer, now what is $\text{prob}(a|b) = 450/500$

So, for $p(b|a) =$ if a already happened that 550 people are tested +ve, but how many people actually have cancer? = $450/550 = 9/11$